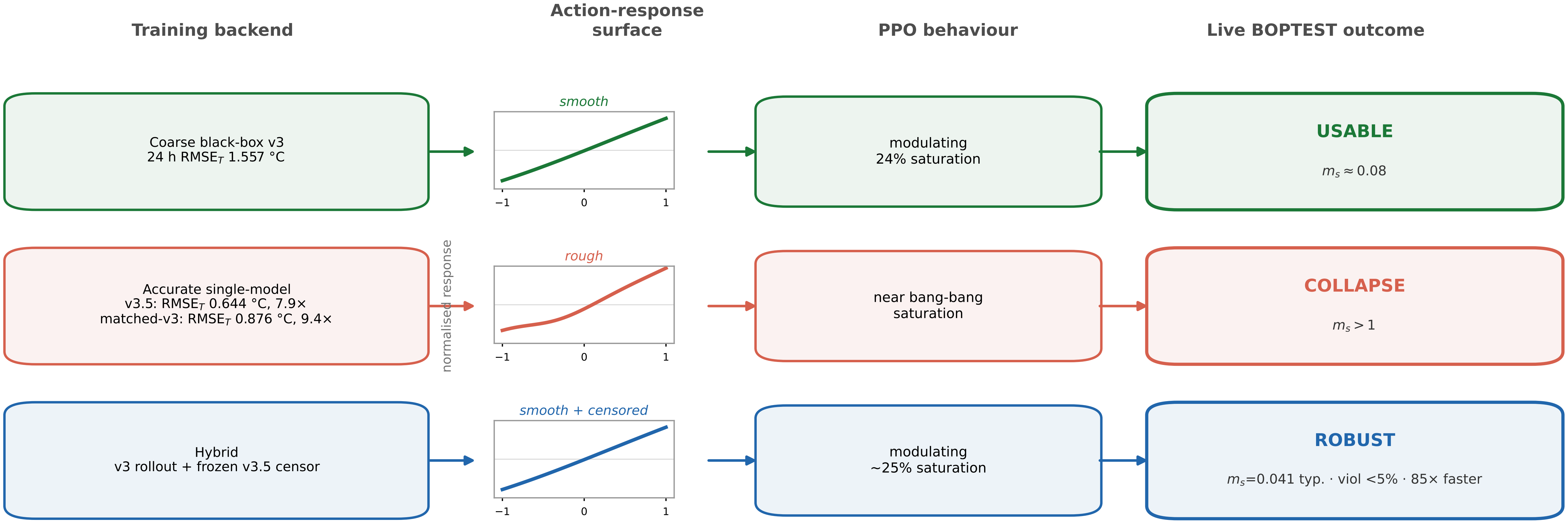
Fidelity–Utility Paradox in Surrogate-Based RL for HVAC Control

More accurate surrogates expose rougher action surfaces and collapse PPO; hybrid separates smooth rollout from physical censoring.



RMSE$_T$ = 24 h predictive temperature error · m_s = live maintenance score (lower is better) · violation = fraction of time outside the 21-24 °C comfort band